

## Appendix I

### 1) Source of Dataset

The dataset used in this study is the Matched Multiple Birth and Fetal Death Data File, 2016–2020, publicly released by the National Center for Health Statistics (NCHS) under the U.S. Centers for Disease Control and Prevention (CDC). This specialized file was designed to enable analysis of matched sets of multiple births and fetal deaths (twins, triplets, and quadruplets) occurring among U.S. residents between 2016 and 2020. It links individual birth and fetal death records from the U.S. *Vital Statistics Natality* and *Fetal Death* files using common identifiers such as plurality, state and county of occurrence, facility ID, and the mother's date of birth.

A total of 641,934 records are included:

- 633,734 live births,
- 8,200 fetal deaths,
- and 7,193 infant deaths linked back to their corresponding birth records.

This linkage allows researchers to study outcomes within a set of multiples (for example, comparing birthweight, survival status, or gestational age between twins). Higher-order multiples (quintuplets or more) are excluded for confidentiality reasons. The public documentation and codebook are available on the CDC Vital Statistics website:

[Data Access - Vital Statistics Online](#)

and detailed variable definitions are at

[https://ftp.cdc.gov/pub/Health\\_Statistics/NCHS/Dataset\\_Documentation/DVS/matched-multiples/matched-multiple-birth-fetal-death-2016-2020.pdf](https://ftp.cdc.gov/pub/Health_Statistics/NCHS/Dataset_Documentation/DVS/matched-multiples/matched-multiple-birth-fetal-death-2016-2020.pdf).

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### 2) Data Dictionary (**Variables Used in This Study**)

The group focused on variables describing birth outcomes, maternal characteristics, and delivery conditions relevant to analyzing perinatal risk factors in multiple births. Below is a summary of the subset of variables used from the full NCHS codebook:

| Variable | Description<br>(Label) | Definition / Coding      | Category   |
|----------|------------------------|--------------------------|------------|
| BIRTHID  | Record category        | 1 = Survivor; 2 = Infant | Outcome ID |

|                |                                       |                                                                                                                     |                       |
|----------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------|
|                |                                       | death; 3 = Fetal death                                                                                              |                       |
| PLURAL         | Type of multiple birth                | 2 = Twin; 3 = Triplet; 4 = Quadruplet                                                                               | Birth type            |
| COUNT          | Number of matched records in set      | 1 = Unmatched; 2 = Two matched; 3 = Three matched; 4 = Four matched                                                 | Match completeness    |
| MULTID         | Matched ID in multiple delivery       | Numeric (100000–799999)                                                                                             | Set identifier        |
| SEX            | Sex of infant                         | M = Male; F = Female                                                                                                | Infant characteristic |
| NO_ABNORM      | No abnormal infant conditions checked | 1 = TRUE; 0 = FALSE; 9 = Not reported                                                                               | Infant health         |
| AGER5          | Infant age at death (recode)          | 1 = <1 hour; 2 = 1–23 hours; 3 = 1–6 days; 4 = 7–27 days; 5 = ≥ 28 days                                             | Mortality timing      |
| BWTR12 / BWTR4 | Birthweight recodes                   | 12-category or 4-category weight bands (227 g – 8165 g)                                                             | Infant health         |
| MRACEHISP      | Maternal race/Hispanic origin         | 1 = Non-Hisp White; 2 = Non-Hisp Black; 3 = AIAN; 4 = Asian; 5 = NHOP; 6 = Multiple race; 7 = Hispanic; 8 = Unknown | Demographics          |
| MAGER9         | Maternal age recode                   | 1 = <15; ... 9 = 50–54                                                                                              | Maternal demographics |
| MEDUC / FEDUC  | Maternal / Paternal education         | 1 = ≤ 8th grade ... 8 = Doctorate; 9 = Unknown                                                                      | Socioeconomic         |
| FAGER11        | Paternal age                          | 1 = <15 ... 11 = Unknown                                                                                            | Demographics          |

| recode                        |                                                  |                                                                          |                    |
|-------------------------------|--------------------------------------------------|--------------------------------------------------------------------------|--------------------|
| NATIVITY                      | Maternal nativity                                | 1 = Born in U.S.; 2 = Born outside U.S.; 3 = Unknown                     | Background         |
| ILP_R11                       | Interval since last pregnancy                    | 0 = 0–3 months ... 8 = ≥ 72 months ... 99 = Unknown                      | Maternal history   |
| RF_CESAR                      | Previous cesarean                                | Y = Yes; N = No; U = Unknown                                             | Maternal condition |
| RF_INFTR                      | Infertility treatment used                       | Y = Yes; N = No; U = Unknown                                             | Maternal condition |
| BMI_R                         | Maternal body mass index (BMI) category          | 1 = Underweight ... 6 = Obesity III (≥ 40); 9 = Unknown                  | Maternal health    |
| CIG_0 – CIG_3 / CIG_R         | Cigarette use before and during pregnancy        | Raw (0–97 cigs/day) and recoded (0 = Nonsmoker ... 5 = 41+)              | Substance use      |
| OEGEST_R12                    | Gestational age (clinician's obstetric estimate) | 1 = <16 weeks ... 11 = 42 weeks or more ... 12 = Unknown                 | Pregnancy course   |
| APGAR5 (APGAR5R)              | 5-minute Apgar score                             | 0–10 score (99 = Unknown)                                                | Infant condition   |
| ME_PRES                       | Fetal presentation at delivery                   | 1 = Cephalic; 2 = Breech; 3 = Other; 9 = Unknown                         | Delivery           |
| ME_ROUT / DMETHOD / DMETHOD_R | Route and method of delivery                     | 1 = Spontaneous ... 4 = Cesarean; Simplified 1 = Vaginal / 2 = C-section | Delivery           |
| NO_MMORB                      | No maternal morbidity                            | Y = Yes; N = No; U = Unknown                                             | Maternal outcome   |

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### 3) Note on Analytical Use

For this project, the variables above will support modeling the association between maternal factors (age, education, BMI, cesarean history, cigarette use, infertility treatment, etc.) and matched birth outcomes (survival, birthweight, gestational age, abnormal conditions).

Continuous variables (e.g., *CIG\_0–CIG\_3*, *OEGEST\_R12*, *BWTR12*) may be analyzed categorically or numerically depending on data distribution.

The matching variables (*MULTID*, *COUNT*, *PLURAL*) enable set-level clustering and intra-pair comparisons among twins and higher-order multiples.